CO3-AT1- EXPERIMENTS

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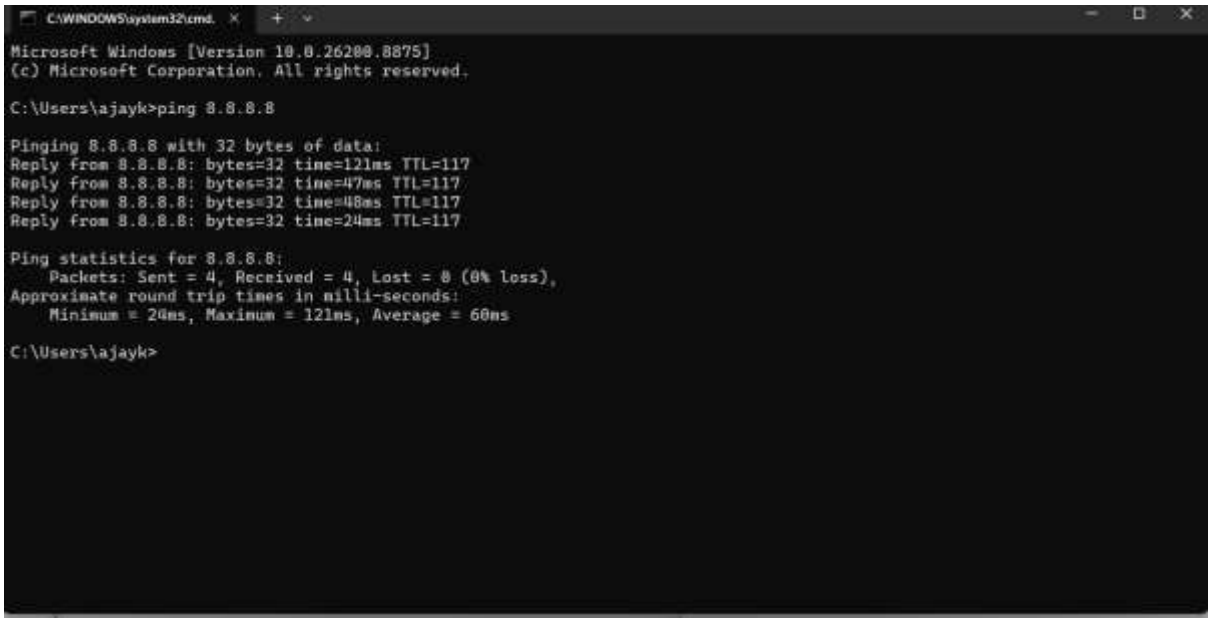
Part 1: Subnetting & IP Design Experiments

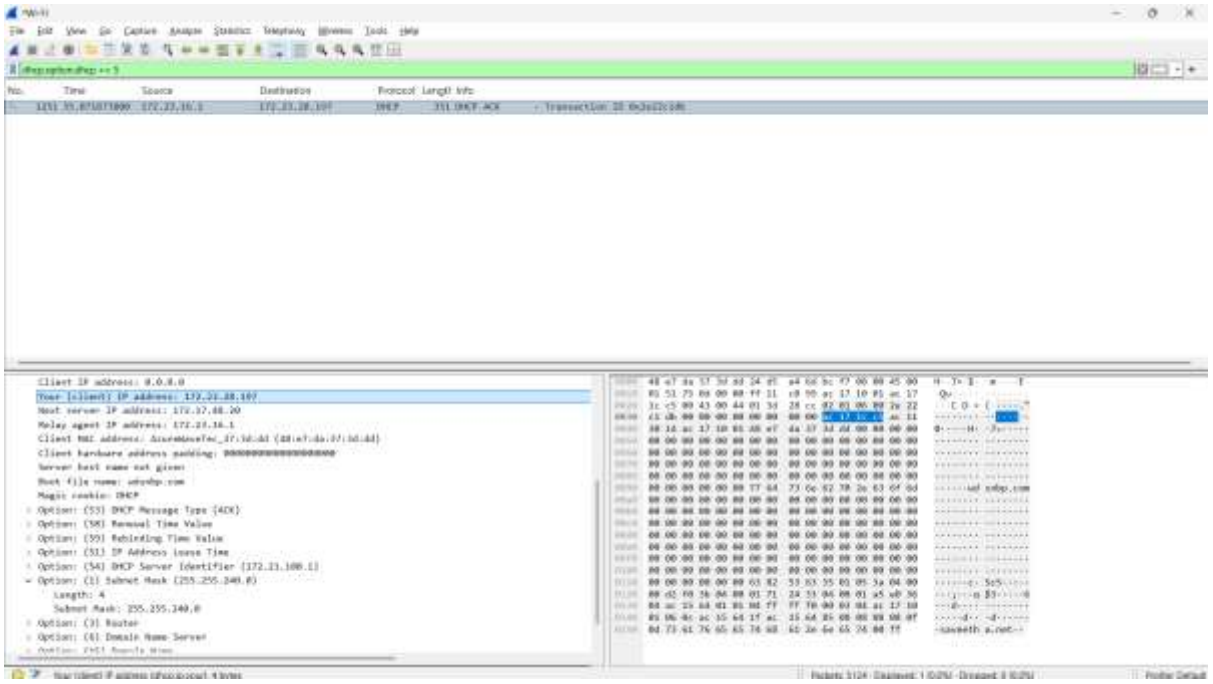
Experiment 2: Subnet Calculation from Host IP

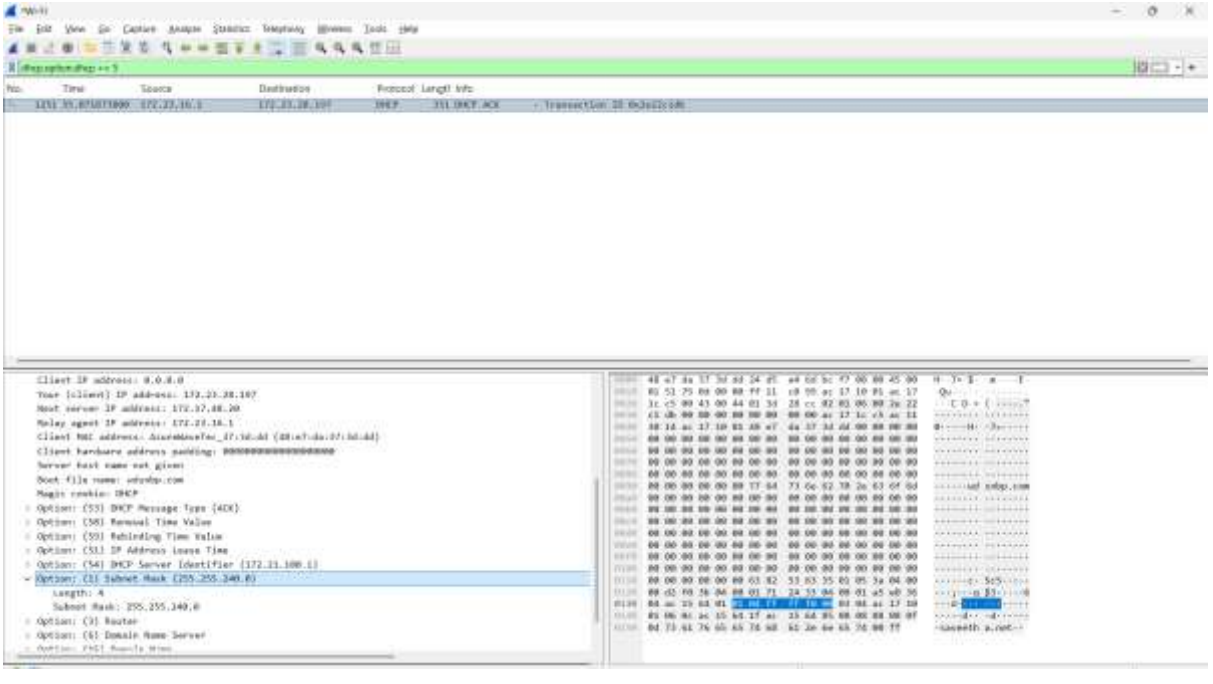
Scenario: During a Wireshark capture experiment, a student notices a workstation sending packets from the IP address 172.16.45.110 with a subnet maskOf 255.255.255.224.

Tasks:Calculate the exact Network Address to which this workstation belongs.Determine the First Usable IP and the Last Usable IP address in this specific subnet.Provide the Broadcast Address for this subnet block.

Output:







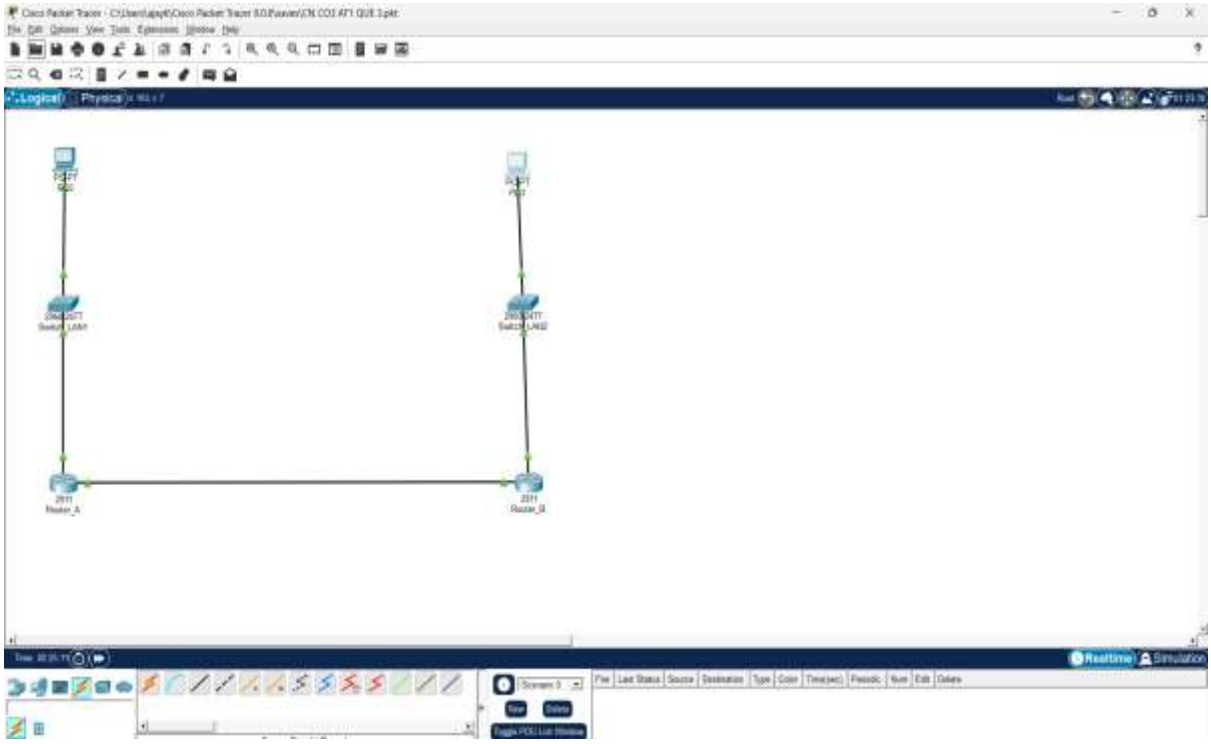
Part 2: Router Configuration & Routing Experiments

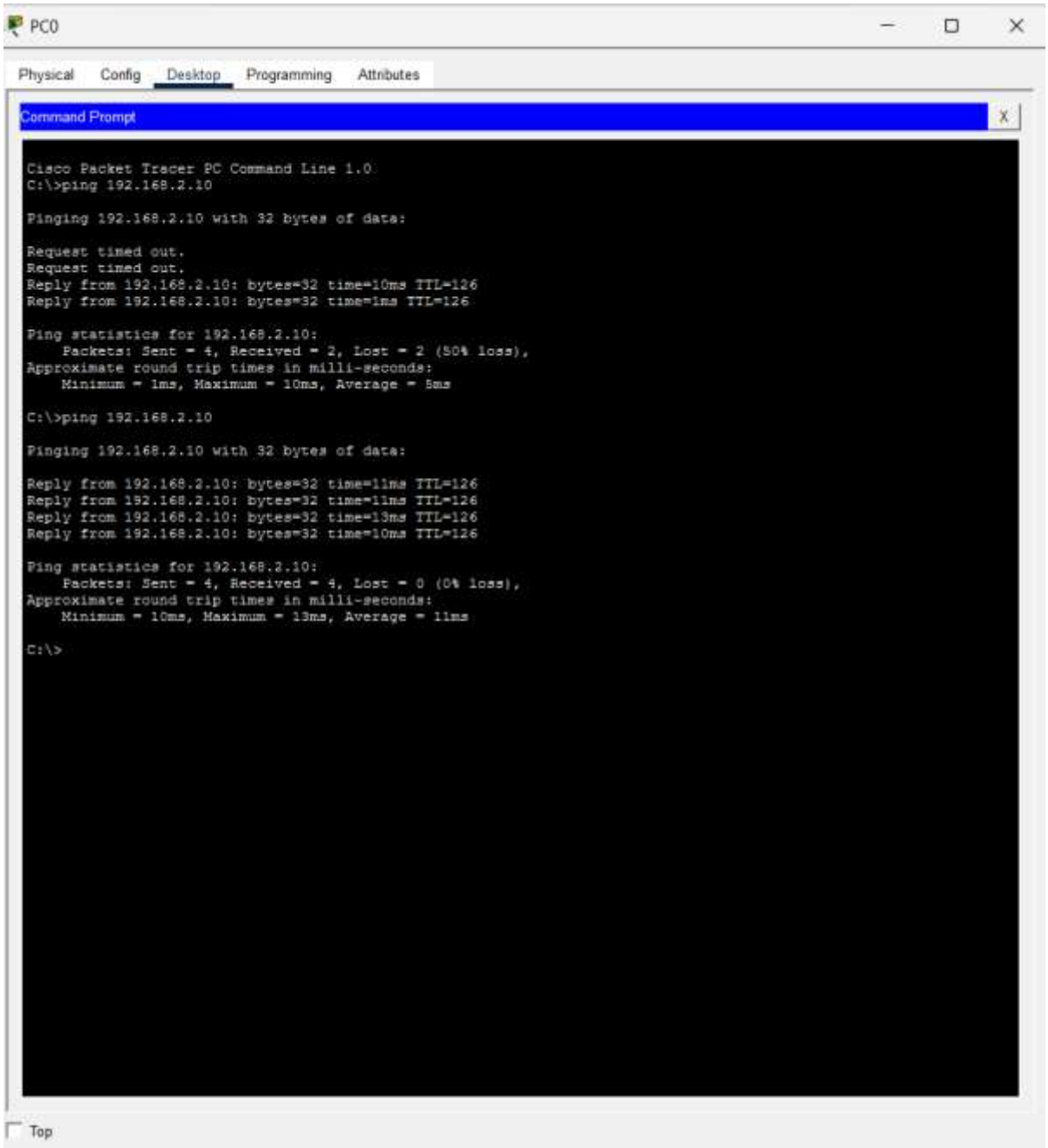
Experiment 3: Cisco Command Line Interface (CLI) Static Routing
Scenario: You are setting up a laboratory topology in Cisco Packet Tracer consisting of two routers (Router_A and Router_B) connected via their GigabitEthernet0/0 interfaces.

- Router_A is connected to Local LAN 1: 192.168.1.0/24
- Router_B is connected to Local LAN 2: 192.168.2.0/24
- The link between the two routers uses the subnet 10.10.10.0/30. Router_A holds 10.10.10.1 and Router_B holds 10.10.10.2.

Tasks:Write out the exact global configuration mode commands required to configure a manual static route on Router_A so it can successfully reach LAN 2.Write the command to configure a default static route on Router_B that points all unknown traffic back toward Router_A.

Output:




PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.2.10: bytes=32 time=10ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 10ms, Average = 5ms
C:\>ping 192.168.2.10
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=11ms TTL=126
Reply from 192.168.2.10: bytes=32 time=11ms TTL=126
Reply from 192.168.2.10: bytes=32 time=13ms TTL=126
Reply from 192.168.2.10: bytes=32 time=10ms TTL=126
Ping statistics for 192.168.2.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 10ms, Maximum = 13ms, Average = 11ms
C:\>
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